Tanisha Srinivasan

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EDUCATION

University of Southern California Expected May 2029
Bachelor of Science in Physics & Computer Science, Minor in Computational Neuroscience *Los Angeles, CA*
- Relevant Coursework: Probability Concepts in Engineering, Mechanics and Thermodynamics

University of California, Santa Cruz June 2025 – June 2026
Computer Science coursework *GPA: 3.90*
- Relevant Coursework: Computer Systems, C Programming & Assembly Language, Linear Algebra, Vector Calculus

EXPERIENCE

Cue, Sigma Eta Pi @ USC Los Angeles, CA
Hardware Engineer *Aug. 2026 – Present*
- Designed a 42 × 32 mm, 4-layer PCB in KiCad for Cue, an ear-worn wearable that gives a silent haptic tap when the wearer uses filler words or speaks too fast.
- Built the board around an nRF54L15 Bluetooth LE module, an nPM1300 PMIC, a PDM mic, a 6-axis IMU, and a DRV2605L haptic driver, reaching ~10 hours of use on a 100 mAh LiPo.
- Verified the 56-component design and had boards fabricated and assembled at JLCPCB; the prototype flags filler words with 90% accuracy across 15 testers.
- Wrote firmware for an nRF52840 prototype that delivers haptic taps within 150 ms of a Bluetooth command from the team's app.

Dexterra Grip, Project Invent San Jose, CA
Hardware Engineer *Jan. 2025 – July 2026*
- Designed and prototyped a $200 assistive glove, about 1/10 the cost of commercial alternatives, providing up to 30 N of proportional grip assistance using an Arduino Nano, force-sensitive resistors, and servo-driven tendon actuation.
- Iterated through 5 actuation concepts, improving grip consistency by 40% based on end-user feedback.
- Named a U.S. finalist in the InventFuture.Global Innovation Challenge; recognized by the U.S. National Innovator Challenge and the Project Invent Moonshot Award.

Tremor-Dampening Device, Pie-Fi Incubator @ UCSC Santa Cruz, CA
Hardware & Embedded Systems Developer *July 2025 – Dec 2025*
- Built a wearable that uses an IMU to detect tremor motion and actuated damping to stabilize the hand, reducing measured tremor amplitude by 35%.
- Designed mechanical components and integrated embedded hardware across 4 prototype iterations.
- Demoed the working prototype at Pie-Fi Demo Day to 100+ attendees.

Exoplanet Detection, Inspirit AI Scholars Santa Cruz, CA
Machine Learning Researcher *Aug. 2024 – Oct. 2024*
- Built convolutional neural networks in Python with TensorFlow/Keras to classify time-series light curves from NASA Kepler data.
- Trained and evaluated models on 100,000+ stellar observations, reaching about 87% validation accuracy.
- Presented methodology and results at the program-wide research symposium.

OTHER PROJECTS

API Security System, Google Developer Group @ UCSC Santa Cruz, CA
Developer *Sept. 2025 – 2026*
- Co-developed a Python proxy that authenticates API requests and blocks replay attacks to protect API credentials.

SKILLS

Hardware: 4-layer PCB design (KiCad), embedded systems, Nordic nRF (Bluetooth LE), power management, I2C, Arduino, Raspberry Pi, sensor and actuator integration, rapid prototyping

Programming: C, C++, Python, MATLAB, Java, JavaScript

Tools & Libraries: SolidWorks, Git/GitHub, NumPy, Pandas, OpenCV, TensorFlow/Keras